

- IOUYAP: This provides network support for IOU (IOS for UNIX).

Supported images

GNS3 (Dynamips) uses actual device IOS images to run the actual IOS rather than simulating device commands. An example of a device simulator is Cisco Packet Tracer. Only selected versions of IOS, and selected modules, cards, processing engines, port adapters, etc, are supported. A list of images supported by GNS3 is given below. Please remember that only selected devices from the mentioned device series are supported.

Cisco router series	1700, 2600, 3600, 3700, 7200	Cisco PIX firewall	PIX 525
Cisco ASA firewall	ASA 5520	Cisco IDS sensor	IDS 4235/4215
Cisco IOU	IOU is IOS for UNIX. These images can be used with GNS3 for switching.	Juniper router	JunOS series M (JunOS of the Juniper router is based on FreeBSD)

Detailed list of supported hardware

A detailed list of supported hardware is available at <https://community.gns3.com/docs/DOC-1708>.

An interesting guideline mentioned in this link is to use older images supporting your configuration requirement, for better speed.

System requirements

To run real IOS image(s) in virtual environment, CPU and RAM resources from your computer are required. Virtualisation also requires additional RAM and CPU capacity.

Considering these requirements, your computer system should have CPU and RAM resources sufficient for the desired IOS image and network topology.

If you wish to configure complex topologies, the best choice will be a 64-bit Linux OS, with a minimum of 4GB RAM, or 64-bit Windows with a minimum of 8GB RAM. This is because, as you must be aware, all 32-bit operating systems have a total addressable space barrier, which limits the maximum RAM limit to 3GB.

Downloading GNS3

Register with gns3.com and proceed to the download section. You may also use the GitHub link <https://github.com/GNS3/gns3-gui/releases>, where you can select the desired version and go ahead. The approximate download size for GNS3 1.3.7 is 65MB for Windows and 40MB for Linux.

Installation instructions

Windows: Download installables and double click.

Ubuntu 14.04: Using the command line, you need to install Python, Dynamips, GNS3 Server, GNS3 GUI, IOU dependencies, IOUYAP, VPCS, Virtual Box, QEMU and cpulimit. You may also consider installing Wireshark to capture packets for diagnostics and troubleshooting.

Step-by-step instructions for installation of GNS3 on Ubuntu 14.04 are available at <https://community.gns3.com/thread/5471>. Complete the installation. GNS3 runs in non-administrative mode, so use the command `gns3`.

Practical uses

GNS3 is a great help for students aspiring for various Cisco certifications. Preparation for most of the laboratory tests for CCNA, CCNP and even CCIE is possible using GNS3.

GNS3 is also a handy tool for professionals. Consider an enterprise with a data centre (DC), disaster recovery (DR) site, primary and redundant links, routers connecting branches to the DC and DR, etc. An incorrect change made to any of the networking devices may lead to incorrect configuration or network outage. Here, GNS3 can be used to configure the virtual network mirroring the actual set-up. This virtual set-up can be used as a test environment to make the desired changes, and then these tested changes can be applied to the live (production) network.

Did you guess an interesting dependency of GNS3, not mentioned above?

Where can Cisco IOS and IOU images be obtained? Are they provided on the GNS3 website? Can anybody provide them?

Let me make it very clear – GNS3 users will have to arrange IOS images on their own. IOS and IOU images are the property of Cisco, and so user licences are available from the company. You have been warned!

Your first lab: GNS3 1.3.7 installed on Ubuntu 14.04

Start GNS3 from the command line as follows:

```
$ gns3
```

During startup, GNS3 will check and inform you if any new updates are available.

You must follow these steps to add your first two routers:

1. Enter the name of the new project. The default project folder is `/home/username/GNS3/projects/`, which can be changed from *Edit - Preferences*. Give your project a meaningful name.
2. Add router images to GNS3.
Go to *Edit - Preferences - Dynamips - IOS Routers -*